
sphinx-mcp

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sphinx-mcp is a [Sphinx](https://www.sphinx-doc.org/)¹ documentation extension for Model Context Protocol (MCP) servers. In this documentation, you will find information on how to install, configure, and use the extension to document MCP servers. You will also find examples of how to use the extension with different MCP servers, such as [pymcp](https://github.com/anirbanbasu/pymcp)², [smt-sudoku-mcp](https://github.com/anirbanbasu/smt-sudoku-mcp)³ and [@modelcontextprotocol/server-everything](https://github.com/modelcontextprotocol/servers/tree/main/src/everything)⁴.

¹ <https://www.sphinx-doc.org/>

² <https://github.com/anirbanbasu/pymcp>

³ <https://github.com/anirbanbasu/smt-sudoku-mcp>

⁴ <https://github.com/modelcontextprotocol/servers/tree/main/src/everything>

CHAPTER

1

USAGE

This section covers the installation, configuration and usage of the `sphinx-mcp` extension for Sphinx.

1.1 Installation

You can install `sphinx-mcp` using `pip` as `pip install sphinx-mcp`. Alternatively, you can add it to your existing project a project management tool, such as `uv`⁵, by calling `uv add sphinx-mcp`. Once installed, you can use it in your Sphinx project by adding it to the `extensions` list in your Sphinx `conf.py` file and add the MCP servers configuration, as shown below.

1.2 Configuration

To document only one MCP server, you can set the `allow_only_one_mcp_server` configuration option to `True` (it defaults to `False` if not specified).

```
extensions = [
    "sphinx_mcp",
]

mcp_config = {
    "mcpServers": {
        "pymcp": {
            "transport": "stdio",
            "command": "uvx",
            "args": ["--from", "pymcp-template", "pymcp"],
        }
    }
}

allow_only_one_mcp_server = True
```

To document multiple MCP servers, you can specify them in the `mcpServers` dictionary. Remember to set the `allow_only_one_mcp_server` to `False` or not set it at all, as it defaults to `False`. In

⁵ <https://docs.astral.sh/uv/>

the following example, three MCP servers are configured: two using `uvx` to run Python packages (`pymcp` and `smt-sudoku-mcp`) without installing them as project dependencies, and another using the `@modelcontextprotocol/server-everything` package, which is not even implemented in Python!

```
extensions = [
    "sphinx_mcp",
]

mcp_config = {
    "mcpServers": {
        "pymcp": {
            # Local stdio server, launched on demand via uvx
            "transport": "stdio",
            "command": "uvx",
            "args": ["--from", "pymcp-template", "pymcp"],
        },
        "smt-sudoku-mcp": {
            "transport": "stdio",
            "command": "uvx",
            "args": ["smt-sudoku-mcp"],
        },
        "everything": {
            # Make sure to run `npm use --lts` before running this.
            "command": "npx",
            "args": ["-y", "@modelcontextprotocol/server-everything"],
        }
    }
}
```

In both the single and multiple server configurations, only `stdio` transport is illustrated. However, servers with `sse` and `streamable-http` transports can also be configured. For `stdio` transport, the command necessary to run the server must execute successfully as Sphinx will use it to connect to the server. This necessitates having the necessary runtime environment set up, such as having the correct Python version or Node.js version installed.

1.3 Directives

The extension provides the following directives to document MCP servers. Each directive belongs to the domain `mcpdocs`.

`mcpdocs::tools`

This directive is used to document the tools available in one or more MCP servers. An optional argument can be provided to specify the MCP server to document. If no argument is provided, it defaults to all the servers configured in the `mcp_config` dictionary. If the argument is absent, each tool listed will be prefixed with the server name, followed by two colons, e.g., `pymcp::greet`.

Arguments:

- server name (optional)

`mcpdocs::prompts`

This directive is used to document the prompts available in one or more MCP servers. Similar to the tools directive, an optional argument can be provided to specify the MCP server to document. If no argument is provided, it defaults to all the servers configured in the `mcp_config` dictionary. If the argument is absent, each prompt listed will be prefixed with the server name, followed by two colons, e.g., `pymcp::code_prompt`.

Arguments:

- server name (optional)

mcpdocs::resources

This directive is used to document the resources available in one or more MCP servers. An optional argument can be provided to specify the MCP server to document. If no argument is provided, it defaults to all the servers configured in the `mcp_config` dictionary. If the argument is absent, each resource listed will be prefixed with the server name, followed by two colons, e.g., `pymcp::resource_logo`.

Arguments:

- server name (optional)

mcpdocs::resource_templates

This directive is used to document the resource templates available in one or more MCP servers. An optional argument can be provided to specify the MCP server to document. If no argument is provided, it defaults to all the servers configured in the `mcp_config` dictionary. If the argument is absent, each resource template listed will be prefixed with the server name, followed by two colons, e.g., `pymcp::resource_unicode_modulo10`.

Arguments:

- server name (optional)

1.4 Limitations

The extension currently only has the aforementioned directives. It does not contain any roles yet. In addition, the generation of indices for the documented tools, prompts, resources, and resource templates is not implemented yet.

CHAPTER

2

EXAMPLE MCP SERVER: PYMCP

`pymcp` is an open-source template repository for developing MCP servers in Python, using [FastMCP](http://gofastmcp.com/)⁶. The project is available on GitHub at [pymcp](https://github.com/anirbanbasu/pymcp)⁷.

2.1 Tools

1. **greet**: Greet the caller with a quintessential Hello World message.

Input schema:

```
{
  "type": "object",
  "additionalProperties": false,
  "properties": {
    "name": {
      "anyOf": [
        {
          "type": "string"
        },
        {
          "type": "null"
        }
      ],
      "default": null,
      "description": "The optional name to be greeted."
    }
  }
}
```

Output schema:

```
{
  "properties": {
```

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⁶ <http://gofastmcp.com/>

⁷ <https://github.com/anirbanbasu/pymcp>

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```
"result": {
  "type": "string"
},
"required": [
  "result"
],
"type": "object",
"x-fastmcp-wrap-result": true
}
```

Annotations:

```
{
  "title": null,
  "readOnlyHint": true,
  "destructiveHint": null,
  "idempotentHint": null,
  "openWorldHint": null
}
```

Metadata:

```
{
  "fastmcp": {
    "tags": [
      "example",
      "greeting"
    ]
  }
}
```

2. **generate_password**: Generate a random password with specified length, optionally including special characters.

Input schema:

```
{
  "type": "object",
  "additionalProperties": false,
  "properties": {
    "length": {
      "default": 12,
      "description": "The length of the password to generate (between 8 and 64↵
↪characters).",
      "maximum": 64,
      "minimum": 8,
      "type": "integer"
    },
    "use_special_chars": {
      "default": false,
      "description": "Include special characters in the password.",
      "type": "boolean"
    }
  }
}
```

Annotations:

```
{
  "title": null,
  "readOnlyHint": true,
  "destructiveHint": null,
  "idempotentHint": null,
  "openWorldHint": null
}
```

Metadata:

```
{
  "fastmcp": {
    "tags": [
      "example",
      "password-generation"
    ]
  }
}
```

3. **text_web_search**: Perform a text web search using the provided query using DDGS.

Input schema:

```
{
  "type": "object",
  "additionalProperties": false,
  "properties": {
    "query": {
      "description": "The search query to fetch results for. It should be a ↵
↵non-empty string.",
      "type": "string"
    },
    "region": {
      "anyOf": [
        {
          "type": "string"
        },
        {
          "type": "null"
        }
      ],
      "default": "uk-en",
      "description": "Optional region to search in."
    },
    "max_results": {
      "anyOf": [
        {
          "maximum": 100,
          "minimum": 1,
          "type": "integer"
        },
        {
          "type": "null"
        }
      ],
      "default": 10,
      "description": "The maximum number of results to return. Default is 10, ↵
↵maximum is 100."
    }
  }
}
```

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```

    },
    "pages": {
      "anyOf": [
        {
          "maximum": 10,
          "minimum": 1,
          "type": "integer"
        },
        {
          "type": "null"
        }
      ],
      "default": 1,
      "description": "The number of pages to fetch. Default is 1, maximum is 10."
    }
  },
  "required": [
    "query"
  ]
}

```

Output schema:

```

{
  "properties": {
    "result": {
      "items": {
        "additionalProperties": true,
        "type": "object"
      },
      "type": "array"
    }
  },
  "required": [
    "result"
  ],
  "type": "object",
  "x-fastmcp-wrap-result": true
}

```

Metadata:

```

{
  "fastmcp": {
    "tags": [
      "meta-search",
      "searchexample",
      "text-search"
    ]
  }
}

```

4. **permutations:** Calculate the number of ways to choose k items from n items without repetition and with order.

Input schema:

```
{
  "type": "object",
  "additionalProperties": false,
  "properties": {
    "n": {
      "description": "The number of items to choose from.",
      "minimum": 1,
      "type": "integer"
    },
    "k": {
      "anyOf": [
        {
          "minimum": 1,
          "type": "integer"
        },
        {
          "type": "null"
        }
      ],
      "default": null,
      "description": "The optional number of items to choose."
    }
  },
  "required": [
    "n"
  ]
}
```

Output schema:

```
{
  "properties": {
    "result": {
      "type": "integer"
    }
  },
  "required": [
    "result"
  ],
  "type": "object",
  "x-fastmcp-wrap-result": true
}
```

Annotations:

```
{
  "title": null,
  "readOnlyHint": true,
  "destructiveHint": null,
  "idempotentHint": null,
  "openWorldHint": null
}
```

Metadata:

```
{
  "fastmcp": {
```

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```
"tags": [  
  "example",  
  "math",  
  "permutation"  
]  
}  
}
```

5. **run_python_code**: Run the given Python code and return the output or error message.

Input schema:

```
{  
  "type": "object",  
  "additionalProperties": false,  
  "properties": {  
    "code": {  
      "type": "string"  
    },  
    "inputs": {  
      "anyOf": [  
        {  
          "additionalProperties": true,  
          "type": "object"  
        },  
        {  
          "type": "null"  
        }  
      ],  
      "default": null  
    },  
    "script_name": {  
      "default": "main.py",  
      "type": "string"  
    },  
    "check_types": {  
      "default": true,  
      "type": "boolean"  
    },  
    "type_definitions": {  
      "anyOf": [  
        {  
          "type": "string"  
        },  
        {  
          "type": "null"  
        }  
      ],  
      "default": null  
    }  
  },  
  "required": [  
    "code"  
  ]  
}
```

Metadata:

```
{
  "fastmcp": {
    "tags": [
      "example",
      "monty",
      "python",
      "secure interpreter"
    ]
  }
}
```

2.2 Prompts

1. **code_prompt**: Get a prompt to write a code snippet in Python based on the specified task.

Input arguments:

```
[
  {
    "name": "task",
    "title": null,
    "description": null,
    "required": true
  }
]
```

Metadata:

```
{
  "fastmcp": {
    "tags": [
      "code-generation",
      "example"
    ]
  }
}
```

2.3 Resources

1. **resource_logo** (data://logo) [application/json]: Get the base64 encoded PNG logo of PyMCP.

Metadata:

```
{
  "fastmcp": {
    "tags": [
      "example",
      "logo",
      "png"
    ]
  }
}
```

2. **resource_logo_svg** (data://logo_svg) [image/svg+xml]: Get the PyMCP logo as SVG.

Metadata:

```
{
  "fastmcp": {
    "tags": [
      "example",
      "logo",
      "svg"
    ]
  }
}
```

2.4 Resource templates

1. **resource_unicode_modulo10** (data://modulo10/{number}) [text/plain]: Computes the modulus 10 of a given number and returns a Unicode character representing the result.

Metadata:

```
{
  "fastmcp": {
    "tags": []
  }
}
```

CHAPTER

3

EXAMPLE MCP SERVER: SMT-SUDOKU-MCP

`smt-sudoku-mcp` is an MCP server that demonstrates the power of satisfiability modulo theories (SMT) solving, using [Z3](https://github.com/Z3Prover/z3)⁸, through the classic constraint-satisfaction puzzle of Sudoku. Unlike `pymcp`, it exposes tools only, with no prompts, resources or resource templates, so this example shows the `mcpdocs::tools` directive on its own.

The project is available on GitHub at [smt-sudoku-mcp](https://github.com/anirbanbasu/smt-sudoku-mcp)⁹.

3.1 Tools

1. **generate_sudoku_puzzle**: Generate a new, uniquely-solvable Sudoku puzzle at the given difficulty.

Input schema:

```
{
  "type": "object",
  "additionalProperties": false,
  "properties": {
    "difficulty": {
      "default": "medium",
      "enum": [
        "very easy",
        "easy",
        "medium",
        "hard",
        "very hard"
      ],
      "type": "string"
    }
  }
}
```

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⁸ <https://github.com/Z3Prover/z3>

⁹ <https://github.com/anirbanbasu/smt-sudoku-mcp>

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```
}  
}
```

Output schema:

```
{  
  "description": "Result of generating a new puzzle.",  
  "properties": {  
    "puzzle": {  
      "description": "A 9x9 Sudoku grid; 0 marks an empty cell.",  
      "properties": {  
        "rows": {  
          "description": "9 rows of 9 cells each; 1-9 for a digit, 0 for empty",  
          "items": {  
            "items": {  
              "type": "integer"  
            },  
            "type": "array"  
          },  
          "type": "array"  
        },  
        "required": [  
          "rows"  
        ],  
        "type": "object"  
      },  
      "difficulty": {  
        "enum": [  
          "very easy",  
          "easy",  
          "medium",  
          "hard",  
          "very hard"  
        ],  
        "type": "string"  
      },  
      "givens": {  
        "description": "Actual number of filled cells in the generated puzzle",  
        "type": "integer"  
      }  
    },  
    "required": [  
      "puzzle",  
      "difficulty",  
      "givens"  
    ],  
    "type": "object"  
  }  
}
```

Metadata:

```
{  
  "fastmcp": {  
    "tags": []  
  }  
}
```

2. **validate_partial_sudoku_solution**: Check whether a partially-filled Sudoku grid is conflict-free and still completable.

Input schema:

```
{
  "type": "object",
  "additionalProperties": false,
  "properties": {
    "grid": {
      "description": "A 9x9 Sudoku grid; 0 marks an empty cell.",
      "properties": {
        "rows": {
          "description": "9 rows of 9 cells each; 1-9 for a digit, 0 for empty",
          "items": {
            "items": {
              "type": "integer"
            },
            "type": "array"
          },
          "type": "array"
        }
      },
      "required": [
        "rows"
      ],
      "type": "object"
    }
  },
  "required": [
    "grid"
  ]
}
```

Output schema:

```
{
  "description": "Result of validating a partially-filled grid.",
  "properties": {
    "conflicts": {
      "items": {
        "description": "A single cell's position.\n\nrow and col are 1-indexed,␣
→ matching how Sudoku cells are conventionally described in text,␣
→ the 0-indexed positions used internally.",
        "properties": {
          "row": {
            "description": "1-indexed row",
            "maximum": 9,
            "minimum": 1,
            "type": "integer"
          },
          "col": {
            "description": "1-indexed column",
            "maximum": 9,
            "minimum": 1,
            "type": "integer"
          }
        }
      }
    }
  },
}
```

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```

        "required": [
            "row",
            "col"
        ],
        "type": "object"
    },
    "type": "array"
},
"is_completable": {
    "anyOf": [
        {
            "type": "boolean"
        },
        {
            "type": "null"
        }
    ],
    "description": "Whether the grid can still be completed; None when
→has_conflicts is True, since completability is not a meaningful question until
→conflicts are resolved"
},
"empty_cells": {
    "description": "Every still-empty cell in the grid, regardless of
→has_conflicts",
    "items": {
        "description": "A single cell's position.\n\nrow and col are 1-indexed,
→matching how Sudoku cells are conventionally described in text,\nrather than
→the 0-indexed positions used internally.",
        "properties": {
            "row": {
                "description": "1-indexed row",
                "maximum": 9,
                "minimum": 1,
                "type": "integer"
            },
            "col": {
                "description": "1-indexed column",
                "maximum": 9,
                "minimum": 1,
                "type": "integer"
            }
        },
        "required": [
            "row",
            "col"
        ],
        "type": "object"
    },
    "type": "array"
},
"has_conflicts": {
    "readOnly": true,
    "type": "boolean"
},
"empty_cells_count": {
    "description": "Number of still-empty cells, i.e. len(empty_cells)",

```

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```

        "readOnly": true,
        "type": "integer"
    }
},
"required": [
    "conflicts",
    "is_completable",
    "empty_cells",
    "has_conflicts",
    "empty_cells_count"
],
"type": "object"
}

```

Metadata:

```

{
  "fastmcp": {
    "tags": []
  }
}

```

3. **validate_full_sudoku_solution:** Check whether a fully-filled Sudoku grid is a correct solution.

Input schema:

```

{
  "type": "object",
  "additionalProperties": false,
  "properties": {
    "grid": {
      "description": "A 9x9 Sudoku grid; 0 marks an empty cell.",
      "properties": {
        "rows": {
          "description": "9 rows of 9 cells each; 1-9 for a digit, 0 for empty",
          "items": {
            "items": {
              "type": "integer"
            },
            "type": "array"
          },
          "type": "array"
        }
      },
      "required": [
        "rows"
      ],
      "type": "object"
    }
  },
  "required": [
    "grid"
  ]
}

```

Output schema:

```
{
  "description": "Result of validating a fully-filled grid.",
  "properties": {
    "has_empty_cells": {
      "type": "boolean"
    },
    "conflicts": {
      "items": {
        "description": "A single cell's position.\n\nrow and col are 1-indexed,
↪ matching how Sudoku cells are conventionally described in text,\nrather than
↪ the 0-indexed positions used internally.",
        "properties": {
          "row": {
            "description": "1-indexed row",
            "maximum": 9,
            "minimum": 1,
            "type": "integer"
          },
          "col": {
            "description": "1-indexed column",
            "maximum": 9,
            "minimum": 1,
            "type": "integer"
          }
        },
        "required": [
          "row",
          "col"
        ],
        "type": "object"
      },
      "type": "array"
    },
    "is_valid": {
      "readOnly": true,
      "type": "boolean"
    }
  },
  "required": [
    "has_empty_cells",
    "conflicts",
    "is_valid"
  ],
  "type": "object"
}
```

Metadata:

```
{
  "fastmcp": {
    "tags": []
  }
}
```

4. **solve_sudoku_puzzle**: Solve an unsolved Sudoku grid, or report why it cannot be solved.

Input schema:

```
{
  "type": "object",
  "additionalProperties": false,
  "properties": {
    "grid": {
      "description": "A 9x9 Sudoku grid; 0 marks an empty cell.",
      "properties": {
        "rows": {
          "description": "9 rows of 9 cells each; 1-9 for a digit, 0 for empty",
          "items": {
            "items": {
              "type": "integer"
            },
            "type": "array"
          },
          "type": "array"
        }
      },
      "required": [
        "rows"
      ],
      "type": "object"
    }
  },
  "required": [
    "grid"
  ]
}
```

Output schema:

```
{
  "description": "Result of solving an unsolved grid.",
  "properties": {
    "status": {
      "enum": [
        "satisfiable",
        "conflicting_givens",
        "unsatisfiable"
      ],
      "type": "string"
    },
    "solution": {
      "anyOf": [
        {
          "description": "A 9x9 Sudoku grid; 0 marks an empty cell.",
          "properties": {
            "rows": {
              "description": "9 rows of 9 cells each; 1-9 for a digit, 0 for empty",
              "items": {
                "items": {
                  "type": "integer"
                },
                "type": "array"
              },
              "type": "array"
            }
          }
        }
      ]
    }
  }
}
```

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```

        }
      },
      "required": [
        "rows"
      ],
      "type": "object"
    },
    {
      "type": "null"
    }
  ]
},
"conflicts": {
  "description": "Populated only when status is conflicting_givens",
  "items": {
    "description": "A single cell's position.\n\nrow and col are 1-indexed,
→ matching how Sudoku cells are conventionally described in text,\nrather than
→ the 0-indexed positions used internally.",
    "properties": {
      "row": {
        "description": "1-indexed row",
        "maximum": 9,
        "minimum": 1,
        "type": "integer"
      },
      "col": {
        "description": "1-indexed column",
        "maximum": 9,
        "minimum": 1,
        "type": "integer"
      }
    },
    "required": [
      "row",
      "col"
    ],
    "type": "object"
  },
  "type": "array"
}
},
"required": [
  "status",
  "solution",
  "conflicts"
],
"type": "object"
}

```

Metadata:

```

{
  "fastmcp": {
    "tags": []
  }
}

```

CHAPTER

4

EXAMPLE MCP SERVER: EVERYTHING

everything is a **Model Context Protocol (MCP) server** that

attempts to exercise all the features of the MCP protocol. It is not intended to be a useful server, but rather a test server for builders of MCP clients. It implements prompts, tools, resources, sampling, and more to showcase MCP capabilities.

Source: [everything](https://github.com/modelcontextprotocol/servers/tree/main/src/everything)¹⁰ as part of the official MCP examples on GitHub.

4.1 Tools

1. **echo**: Echoes back the input string

Input schema:

```
{
  "$schema": "http://json-schema.org/draft-07/schema#",
  "type": "object",
  "properties": {
    "message": {
      "type": "string",
      "description": "Message to echo"
    }
  },
  "required": [
    "message"
  ]
}
```

Annotations:

```
{
  "title": null,
  "readOnlyHint": true,
  "destructiveHint": false,
```

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¹⁰ <https://github.com/modelcontextprotocol/servers/tree/main/src/everything>

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```

    "idempotentHint": true,
    "openWorldHint": false
  }

```

2. **get-annotated-message:** Demonstrates how annotations can be used to provide metadata about content.

Input schema:

```

{
  "$schema": "http://json-schema.org/draft-07/schema#",
  "type": "object",
  "properties": {
    "messageType": {
      "type": "string",
      "enum": [
        "error",
        "success",
        "debug"
      ],
      "description": "Type of message to demonstrate different annotation",
      "patterns":
    },
    "includeImage": {
      "default": false,
      "description": "Whether to include an example image",
      "type": "boolean"
    }
  },
  "required": [
    "messageType"
  ]
}

```

Annotations:

```

{
  "title": null,
  "readOnlyHint": true,
  "destructiveHint": false,
  "idempotentHint": true,
  "openWorldHint": false
}

```

3. **get-env:** Returns all environment variables, helpful for debugging MCP server configuration

Input schema:

```

{
  "$schema": "http://json-schema.org/draft-07/schema#",
  "type": "object",
  "properties": {}
}

```

Annotations:

```

{
  "title": null,

```

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```
"readOnlyHint": true,  
"destructiveHint": false,  
"idempotentHint": true,  
"openWorldHint": false  
}
```

4. **get-resource-links**: Returns up to ten resource links that reference different types of resources

Input schema:

```
{  
  "$schema": "http://json-schema.org/draft-07/schema#",  
  "type": "object",  
  "properties": {  
    "count": {  
      "default": 3,  
      "description": "Number of resource links to return (1-10)",  
      "type": "number",  
      "minimum": 1,  
      "maximum": 10  
    }  
  }  
}
```

Annotations:

```
{  
  "title": null,  
  "readOnlyHint": true,  
  "destructiveHint": false,  
  "idempotentHint": true,  
  "openWorldHint": false  
}
```

5. **get-resource-reference**: Returns a resource reference that can be used by MCP clients

Input schema:

```
{  
  "$schema": "http://json-schema.org/draft-07/schema#",  
  "type": "object",  
  "properties": {  
    "resourceType": {  
      "default": "Text",  
      "type": "string",  
      "enum": [  
        "Text",  
        "Blob"  
      ]  
    },  
    "resourceId": {  
      "default": 1,  
      "description": "ID of the text resource to fetch",  
      "type": "number"  
    }  
  }  
}
```

Annotations:

```
{
  "title": null,
  "readOnlyHint": true,
  "destructiveHint": false,
  "idempotentHint": true,
  "openWorldHint": false
}
```

6. **get-structured-content:** Returns structured content along with an output schema for client data validation

Input schema:

```
{
  "$schema": "http://json-schema.org/draft-07/schema#",
  "type": "object",
  "properties": {
    "location": {
      "type": "string",
      "enum": [
        "New York",
        "Chicago",
        "Los Angeles"
      ],
      "description": "Choose city"
    }
  },
  "required": [
    "location"
  ]
}
```

Output schema:

```
{
  "$schema": "http://json-schema.org/draft-07/schema#",
  "type": "object",
  "properties": {
    "temperature": {
      "type": "number",
      "description": "Temperature in celsius"
    },
    "conditions": {
      "type": "string",
      "description": "Weather conditions description"
    },
    "humidity": {
      "type": "number",
      "description": "Humidity percentage"
    }
  },
  "required": [
    "temperature",
    "conditions",
    "humidity"
  ],
}
```

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```

    "additionalProperties": false
  }

```

Annotations:

```

{
  "title": null,
  "readOnlyHint": true,
  "destructiveHint": false,
  "idempotentHint": true,
  "openWorldHint": false
}

```

7. **get-sum**: Returns the sum of two numbers

Input schema:

```

{
  "$schema": "http://json-schema.org/draft-07/schema#",
  "type": "object",
  "properties": {
    "a": {
      "type": "number",
      "description": "First number"
    },
    "b": {
      "type": "number",
      "description": "Second number"
    }
  },
  "required": [
    "a",
    "b"
  ]
}

```

Annotations:

```

{
  "title": null,
  "readOnlyHint": true,
  "destructiveHint": false,
  "idempotentHint": true,
  "openWorldHint": false
}

```

8. **get-tiny-image**: Returns a tiny MCP logo image.

Input schema:

```

{
  "$schema": "http://json-schema.org/draft-07/schema#",
  "type": "object",
  "properties": {}
}

```

Annotations:

```
{
  "title": null,
  "readOnlyHint": true,
  "destructiveHint": false,
  "idempotentHint": true,
  "openWorldHint": false
}
```

9. **gzip-file-as-resource**: Compresses a single file using gzip compression. Depending upon the selected output type, returns either the compressed data as a gzipped resource or a resource link, allowing it to be downloaded in a subsequent request during the current session.

Input schema:

```
{
  "$schema": "http://json-schema.org/draft-07/schema#",
  "type": "object",
  "properties": {
    "name": {
      "default": "README.md.gz",
      "type": "string",
      "description": "Name of the output file"
    },
    "data": {
      "default":
→ "https://raw.githubusercontent.com/modelcontextprotocol/servers/refs/heads/main/README.md
→",
      "type": "string",
      "format": "uri",
      "description": "URL or data URI of the file content to compress"
    },
    "outputType": {
      "default": "resourceLink",
      "description": "How the resulting gzipped file should be returned.
→ 'resourceLink' returns a link to a resource that can be read later, 'resource'
→ returns a full resource object.",
      "type": "string",
      "enum": [
        "resourceLink",
        "resource"
      ]
    }
  }
}
```

Annotations:

```
{
  "title": null,
  "readOnlyHint": false,
  "destructiveHint": false,
  "idempotentHint": true,
  "openWorldHint": true
}
```

10. **toggle-simulated-logging**: Toggles simulated, random-leveled logging on or off.

Input schema:

```
{
  "$schema": "http://json-schema.org/draft-07/schema#",
  "type": "object",
  "properties": {}
}
```

Annotations:

```
{
  "title": null,
  "readOnlyHint": false,
  "destructiveHint": false,
  "idempotentHint": false,
  "openWorldHint": false
}
```

11. **toggle-subscriber-updates:** Toggles simulated resource subscription updates on or off.

Input schema:

```
{
  "$schema": "http://json-schema.org/draft-07/schema#",
  "type": "object",
  "properties": {}
}
```

Annotations:

```
{
  "title": null,
  "readOnlyHint": false,
  "destructiveHint": false,
  "idempotentHint": false,
  "openWorldHint": false
}
```

12. **trigger-long-running-operation:** Demonstrates a long running operation with progress updates.

Input schema:

```
{
  "$schema": "http://json-schema.org/draft-07/schema#",
  "type": "object",
  "properties": {
    "duration": {
      "default": 10,
      "description": "Duration of the operation in seconds",
      "type": "number"
    },
    "steps": {
      "default": 5,
      "description": "Number of steps in the operation",
      "type": "number"
    }
  }
}
```

Annotations:

```
{
  "title": null,
  "readOnlyHint": true,
  "destructiveHint": false,
  "idempotentHint": true,
  "openWorldHint": false
}
```

13. **simulate-research-query**: Simulates a deep research operation that gathers, analyzes, and synthesizes information. Demonstrates MCP task-based operations with progress through multiple stages. If 'ambiguous' is true and client supports elicitation, sends an elicitation request for clarification.

Input schema:

```
{
  "$schema": "http://json-schema.org/draft-07/schema#",
  "type": "object",
  "properties": {
    "topic": {
      "type": "string",
      "description": "The research topic to investigate"
    },
    "ambiguous": {
      "default": false,
      "description": "Simulate an ambiguous query that requires clarification",
      →(triggers input_required status),
      "type": "boolean"
    }
  },
  "required": [
    "topic"
  ]
}
```

Annotations:

```
{
  "title": null,
  "readOnlyHint": false,
  "destructiveHint": false,
  "idempotentHint": false,
  "openWorldHint": false
}
```

4.2 Prompts

1. **simple-prompt**: A prompt with no arguments
2. **args-prompt**: A prompt with two arguments, one required and one optional

Input arguments:

```
[
  {
    "name": "city",
    "title": null,
```

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```

    "description": "Name of the city",
    "required": true
  },
  {
    "name": "state",
    "title": null,
    "description": null,
    "required": false
  }
]

```

3. **completable-prompt:** First argument choice narrows values for second argument.

Input arguments:

```

[
  {
    "name": "department",
    "title": null,
    "description": "Choose the department.",
    "required": true
  },
  {
    "name": "name",
    "title": null,
    "description": "Choose a team member to lead the selected department.",
    "required": true
  }
]

```

4. **resource-prompt:** A prompt that includes an embedded resource reference

Input arguments:

```

[
  {
    "name": "resourceType",
    "title": null,
    "description": "Type of resource to fetch",
    "required": true
  },
  {
    "name": "resourceId",
    "title": null,
    "description": "ID of the text resource to fetch",
    "required": true
  }
]

```

4.3 Resources

1. **architecture.md** (demo://resource/static/document/architecture.md) [text/markdown]: Static document file exposed from /docs: architecture.md
2. **extension.md** (demo://resource/static/document/extension.md) [text/markdown]: Static document file exposed from /docs: extension.md

3. **features.md** (demo://resource/static/document/features.md) [text/markdown]: Static document file exposed from /docs: features.md
4. **how-it-works.md** (demo://resource/static/document/how-it-works.md) [text/markdown]: Static document file exposed from /docs: how-it-works.md
5. **instructions.md** (demo://resource/static/document/instructions.md) [text/markdown]: Static document file exposed from /docs: instructions.md
6. **startup.md** (demo://resource/static/document/startup.md) [text/markdown]: Static document file exposed from /docs: startup.md
7. **structure.md** (demo://resource/static/document/structure.md) [text/markdown]: Static document file exposed from /docs: structure.md

4.4 Resource templates

1. **Dynamic Text Resource** (demo://resource/dynamic/text/{resourceId}) [text/plain]: Plaintext dynamic resource fabricated from the {resourceId} variable, which must be an integer.
2. **Dynamic Blob Resource** (demo://resource/dynamic/blob/{resourceId}) [application/octet-stream]: Binary (base64) dynamic resource fabricated from the {resourceId} variable, which must be an integer.